CONSULTING CASE STUDY

Retail Inventory & Sales Optimization

Supermarket Chain • 3 Branches • Q1 2019

Tool: MySQL • Window Functions • Inventory Turnover Analysis

$322,967 Total Revenue	**$110,569** Top Branch (C)	**5.58** Best Turnover (C)	**$56,145** Top Product Line

The Business Problem

A three-branch supermarket chain operating across Mandalay, Naypyitaw, and Yangon needed to answer a question its management team could not resolve from raw transaction data alone: which branches and product lines deserve more investment, and which are underperforming relative to their potential?

The chain had 1,000 transactions across Q1 2019 but no structured analysis of branch efficiency, product line prioritization, or inventory utilization. Decisions about stock allocation and marketing spend were being made without data.

> *Engagement objective: Identify the highest-value branches and product lines, quantify inventory efficiency by location, and deliver prioritized recommendations for stock allocation and sales strategy.*

Scope of Analysis

- Branch performance: total revenue and ranking across 3 locations
- Product line analysis: revenue contribution across 6 categories
- Inventory turnover: efficiency ratio per branch
- Regional trend: monthly sales patterns across 3 cities
- Cross-branch product performance: where each category performs strongest

Branch Performance Analysis

Total Revenue by Branch

Branch C leads the chain with $110,568.86 in total Q1 revenue — a 4.1% premium over Branch A and Branch B, which are statistically level at $106,200 and $106,198 respectively. The near-identical performance of A and B signals similar demand environments, while Branch C's outperformance is attributable to its stronger Food and Beverages category.

Branch	City	Total Revenue	Rank	vs Chain Average
C	Naypyitaw	$110,568.86	1st	+4.1% above avg
A	Yangon	$106,200.57	2nd	-0.3% below avg
B	Mandalay	$106,198.00	3rd	-0.3% below avg

Inventory Turnover by Branch

Inventory turnover measures how efficiently a branch converts stock into sales. Branch C again leads at 5.58, meaning it cycles through inventory 5.58 times per invoice on average. Branch B follows at 5.48 and Branch A at 5.47. The gap between top and bottom is narrow — 0.11 — suggesting the chain operates with broadly consistent inventory discipline, with Branch C holding a marginal edge.

Branch	Inventory Turnover Ratio	Assessment
C	5.5823	Most efficient — highest sales velocity
B	5.4819	Mid-tier — slightly below Branch C
A	5.4676	Lowest — marginally less efficient stock cycling

Branch C's combination of highest revenue AND highest turnover makes it the benchmark location. Its product mix and stock management approach should be studied and replicated across Branches A and B.

Product Line Performance

Revenue by Product Line — Chain Total

Food and Beverages is the top-performing category at $56,144.96, representing 17.4% of total chain revenue. Sports and Travel and Electronic Accessories follow closely. Health and Beauty is the weakest performer at $49,193.84 — a $6,951 gap from the top category — suggesting either weaker demand or inadequate shelf space and promotion in this segment.

Product Line	Total Revenue	% of Chain Revenue	Assessment
Food and Beverages	$56,144.96	17.4%	Priority category — protect stock
Sports and Travel	$55,123.00	17.1%	Strong — consistent across branches
Electronic Accessories	$54,337.64	16.8%	Solid — Branch C opportunity
Fashion Accessories	$54,306.03	16.8%	Stable — low variance
Home and Lifestyle	$53,861.96	16.7%	Steady — Branch A strength
Health and Beauty	$49,193.84	15.2%	Underperforming — review strategy

Branch-Level Product Strengths

Each branch has a distinct product strength that should inform individual stock and promotion strategies:

Branch	Strongest Category	Revenue	Action
A	Home and Lifestyle	$22,417.21	Increase shelf space and promotional activity
B	Health and Beauty	$19,980.70	Only branch where H&B performs — study why
C	Food and Beverages	$23,766.88	Protect stock — highest single category revenue

Regional Sales Trends

Monthly Revenue by City — Q1 2019

All three cities show distinct patterns across January, February, and March 2019. Naypyitaw started strongest in January at $40,434.76 but experienced the sharpest February decline before recovering in March. Mandalay showed the most consistent performance. Yangon dipped significantly in February before recovering.

City	January	February	March	Pattern
Naypyitaw	$40,434.76	$32,934.99	$37,199.11	Sharp Feb dip — seasonal or operational
Yangon	$38,681.19	$29,860.20	$37,659.18	Largest Feb decline — investigate cause
Mandalay	$37,176.16	$34,424.39	$34,597.45	Most consistent — stable demand base

The February dip across all three cities suggests a system-wide demand pattern — likely post-January slowdown common in retail. Targeted February promotions could flatten this seasonal trough and improve Q1 revenue by an estimated 8-12%.

Strategic Recommendations

Five recommendations prioritized by expected business impact. Each is directly supported by findings from the analysis.

1

Replicate Branch C's product mix in Branches A and B

Branch C outperforms on both revenue and inventory turnover. Its Food and Beverages dominance ($23,766 vs $17,163 in Branch A) suggests stock allocation differences. Aligning A and B's F&B inventory to C's levels is estimated to recover $6,000-$9,000 in annual revenue per branch.

2

Investigate and address Health and Beauty underperformance

H&B is the only category below $50,000 chain-wide — a $6,951 gap from the top category. Branch B is the only location where it performs strongly ($19,980). Studying Branch B's H&B approach — shelf placement, promotions, stock levels — and applying it to A and C is the fastest path to closing this gap.

3

Launch February retention promotions to flatten the seasonal dip

All three cities show a February revenue decline averaging 12-15% from January. A targeted mid-January campaign — bundle offers, loyalty discounts, category promotions — timed to sustain February demand could recover an estimated $8,000-$15,000 across the chain annually.

4

Protect Food and Beverages stock levels chain-wide

F&B is the top revenue category at $56,144 and is Branch C's dominant driver. Stockouts in this category represent direct revenue loss. Implement minimum stock thresholds and automated reorder triggers for F&B across all three branches.

5

Monitor Branch A and B inventory turnover weekly

Both branches trail Branch C by 0.11-0.12 on turnover ratio. While the gap is small, consistent monitoring will catch any deterioration early. A weekly turnover report by branch and category gives managers the visibility to act before stock issues affect revenue.

Analysis performed using MySQL • Window Functions • Aggregation & JOIN Queries
Dataset: Supermarket Sales • 1,000 Transactions • Q1 2019 • 3 Branches